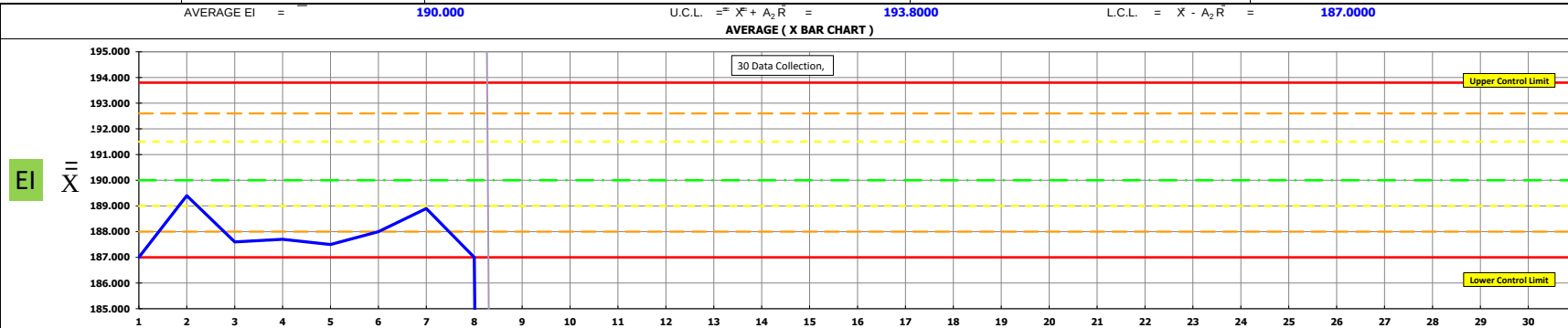


X̄ - R CONTROL CHART

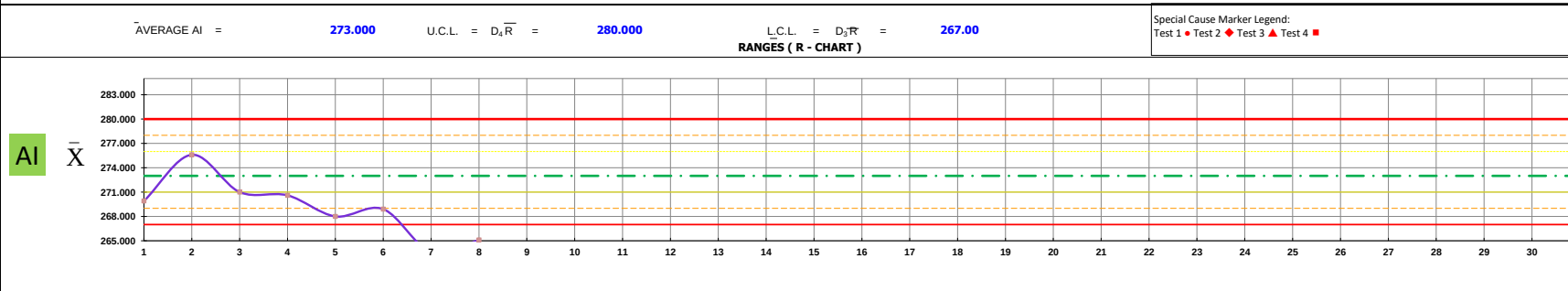
MACHINE M-Tan	SECTION QC	OPERATION M-Tan	DATE CONTROL LIMITS CALC.	SPECIFICATION min EI - : 180 AI Min : 260	COURSE NO.	CONTROL ITEM (Δ) YES NO	☐
	MONTH Jul-25	CHARACTERISTIC Temperatur		SAMPLE SIZE / FREQUENCY n = 3	TYPE / SIZE : D21DW4-B-2 / 0.70		



SPECIAL CAUSES

- ✓ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ✓ TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE
- ✓ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ✓ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ✓ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ✓ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ✓ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ✓ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION	
1.	_____
2.	_____
3.	_____
4.	_____
5.	_____



SUBGROUP SIZE	A_2	d_2	D_2	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

[illegible]

CALCULATION			
$EI(X) = 187.9$ $AI(X) = 269.1$ $AI(StDev) = 3.7 \quad StDev = 0.859$			
THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED			
$C_{PK} = \min \left[\frac{X - LSL}{3\sigma}, \frac{USL - X}{3\sigma} \right]$ $= \min \left[\frac{0.34}{2.29}, \frac{2.29}{1.0} \right] \quad 1.0 \quad 1.0$ $= \underline{0.34} \quad \underline{1.0}$			
$C_p = \frac{USL - LSL}{6\sigma}$ $= \frac{EI}{AI} = \frac{1.32}{0.58}$			
CHECKED BY		APPROVED BY	